

TOMAHAWK, WI, USA

WMO: 722332

Lat: 45.469N

Lon: 89.806W

Elev: 453

StdP: 96.00

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54953

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.9	(c) -24.8	(d) -32.0	(e) 0.2	(f) -27.6	(g) -28.4	(h) 0.3	(i) -23.8	(j) 8.2	(k) -10.2	(l) 7.4	(m) -9.9	(n) 0.9	(o) 280	(p) 0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.8	(c) 29.8	(d) 21.1	(e) 27.8	(f) 19.7	(g) 26.5	(h) 19.0	(i) 22.4	(j) 27.2	(k) 21.4	(l) 25.9	(m) 20.4	(n) 24.6	(o) 2.6	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.5	24.9	20.0	15.5	24.0	18.9	14.5	22.9	68.1	27.1	64.4	25.8	60.9	24.6	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 6.4	(c) 5.4	(e) -32.7	(f) 32.2	(g) 2.1	(h) 1.7	(i) -34.2	(j) 33.5	(k) -35.4	(l) 34.4	(m) -36.6	(n) 35.4	(o) -38.1	(p) 36.6		
(a) 7.5	(b) 6.4	(c) 5.4	(e) -32.9	(f) 24.4	(g) 2.0	(h) 1.1	(i) -34.3	(j) 25.2	(k) -35.5	(l) 25.9	(m) -36.6	(n) 26.5	(o) -38.1	(p) 27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-10.3	-9.3	-2.5	4.6	11.6	16.5	18.8	17.3	13.4	6.7	-0.6	-7.5	(6)
(7)		DBStd	11.66	7.33	7.01	7.07	5.40	4.93	3.67	3.34	3.32	4.64	5.01	5.73	6.38	(7)
(8)		HDD10.0	2788	628	542	392	177	39	1	0	0	18	130	319	543	(8)
(9)		HDD18.3	5002	886	775	646	413	216	77	36	61	159	363	568	801	(9)
(10)		CDD10.0	949	0	0	4	14	88	197	273	226	120	27	1	0	(10)
(11)		CDD18.3	121	0	0	0	0	7	23	50	28	12	2	0	0	(11)
(12)		CDH23.3	1386	0	0	1	8	106	256	579	307	117	13	0	0	(12)
(13)		CDH26.7	299	0	0	0	1	22	49	149	56	22	1	0	0	(13)
(14)	Wind	WSAvg	1.9	2.1	2.1	2.1	2.4	2.1	1.7	1.4	1.3	1.5	1.9	2.0	1.9	(14)
(15)	Precipitation	PrecAvg	818	30	25	45	68	91	105	93	99	106	73	49	36	(15)
(16)		PrecMax	1075	81	61	90	134	162	177	202	263	249	139	171	59	(16)
(17)		PrecMin	614	7	5	7	17	37	41	14	40	44	14	6	10	(17)
(18)		PrecStd	110	20	13	25	34	35	43	47	49	55	37	35	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	9.6	20.3	24.2	29.7	30.9	32.4	30.7	29.1	25.2	16.7	6.8	(19)
(20)			MCWB	2.2	5.2	14.0	14.2	19.7	22.7	21.8	21.1	20.6	17.6	10.5	4.6	(20)
(21)		2%	DB	2.5	5.2	14.9	20.4	26.3	27.8	30.1	27.9	26.6	22.0	12.8	3.7	(21)
(22)			MCWB	1.1	2.5	9.7	11.0	16.9	19.5	21.5	19.8	19.0	16.5	9.3	2.4	(22)
(23)		5%	DB	1.2	2.3	10.9	17.4	23.6	26.3	27.9	26.6	24.3	18.4	10.3	2.2	(23)
(24)			MCWB	0.3	0.5	7.1	9.0	15.5	18.7	20.4	19.1	18.0	13.6	7.0	1.3	(24)
(25)		10%	DB	0.0	0.1	7.4	14.7	21.2	24.6	26.6	25.0	22.2	15.3	7.5	0.7	(25)
(26)			MCWB	-0.8	-1.2	3.8	8.2	14.1	17.7	19.4	18.6	16.9	11.3	5.2	0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	5.7	14.6	15.2	21.2	23.2	24.7	23.0	21.9	20.4	12.7	5.6	(27)
(28)			MCDWB	4.3	9.1	19.3	20.6	27.0	28.3	29.6	27.3	26.3	23.1	14.7	6.0	(28)
(29)		2%	WB	1.3	2.9	11.0	13.0	18.9	21.3	22.8	21.8	20.5	17.0	9.9	2.6	(29)
(30)			MCDWB	2.4	5.1	12.8	18.8	23.6	26.4	27.7	25.8	24.2	20.4	13.1	3.3	(30)
(31)		5%	WB	0.3	0.5	6.8	10.5	17.2	20.0	21.6	20.8	19.4	14.3	7.2	1.1	(31)
(32)			MCDWB	1.0	1.9	10.7	15.4	21.5	24.5	26.4	24.6	22.9	17.9	9.6	1.8	(32)
(33)		10%	WB	-0.9	-1.0	4.1	8.4	15.5	18.8	20.5	19.8	17.9	11.9	5.3	0.1	(33)
(34)			MCDWB	-0.2	0.0	6.9	14.0	19.5	22.8	24.9	23.7	21.2	14.6	7.3	0.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.7	11.5	12.3	12.4	13.8	13.1	13.8	13.4	13.3	10.7	8.7	8.3	(35)
(36)			MCDBR	8.0	11.0	14.3	17.9	16.6	14.7	14.9	14.9	15.6	15.6	13.4	8.1	(36)
(37)			MCWBR	6.9	8.3	8.9	10.0	8.5	8.0	7.7	7.9	8.7	9.5	9.3	7.0	(37)
(38)		5% WB	MCDBR	7.6	9.9	12.8	15.5	13.2	12.8	12.9	12.1	11.8	12.6	12.2	6.7	(38)
(39)			MCWBR	6.7	7.6	8.5	9.6	7.4	7.6	7.4	7.0	7.3	8.7	8.9	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.242	0.269	0.308	0.338	0.366	0.366	0.364	0.365	0.343	0.308	0.281	0.250	(40)	
(41)		taud	2.378	2.299	2.355	2.410	2.371	2.391	2.412	2.415	2.479	2.564	2.535	2.437	(41)	
(42)		Ebn at Noon	890	919	925	923	902	900	897	884	878	871	839	842	(42)	
(43)		Edh at Noon	71	95	105	110	118	116	113	108	93	74	63	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.58	2.70	3.87	4.58	5.36	5.83	6.04	5.11	3.99	2.49	1.49	1.10	(44)	
(45)		RadStd	0.16	0.26	0.38	0.45	0.44	0.47	0.30	0.34	0.27	0.26	0.15	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page